



The influence of pay transparency on (gender) inequity, inequality and the performance basis of pay

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Recent decades have witnessed a growing focus on two distinct income patterns: persistent pay inequity, particularly a gender pay gap, and growing pay inequality. Pay transparency is widely advanced as a remedy for both. Yet we know little about the systemic influence of this policy on the evolution of pay practices within organizations. To address this void, we assemble a dataset combining detailed performance, demographic and salary data for approximately 100,000 US academics between 1997 and 2017. We then exploit staggered shocks to wage transparency to explore how this change reshapes pay practices. We find evidence that pay transparency causes significant increases in both the equity and equality of pay, and significant and sizeable reductions in the link between pay and individually measured performance.

Recent decades have witnessed a growing global focus on two distinct income patterns: persistent pay inequity, particularly a gender pay gap, and growing pay inequality^{1,2}. Though these terms are sometimes used rather interchangeably, pay equity refers to the fairness by which pay is allocated, often measured as the consistency or non-discriminatory manner by which pay is matched to performance or effort. By contrast, pay equality is self-evidently the equivalence of pay, often measured as simply the variance in pay within an organization or society³. While recent studies suggest global reductions in the magnitude of still-persistent pay inequity (specifically the gender pay gap^{4,5}), they also consistently point to a global pattern of increasing pay inequality within organizations⁶ and societies⁷.

In partial response to these patterns have come abundant calls from politicians and advocacy groups for greater transparency in pay allocation, particularly the public disclosure of individual income⁸. The argument is that enhanced pay transparency places social pressure on those allocating pay to reduce both inequity and inequality, including by gender. Accordingly, many nations, states and organizations have taken directional steps to heed this call. But resistance to pay transparency within the private sector remains quite deep-seated. A recent survey of US employers suggests that 41% actively discourage their employees from simply sharing information about pay with their organizational peers, while 25% explicitly prohibit it⁹. The common explanation is that the heightened focus on equity and equality that pay transparency prompts undermines the capacity to link individual pay and performance, thereby compromising organizational efforts to effectively motivate employees and attract talented ones.

Some prior but mostly contemporaneous research explores pay transparency's influence on specific dimensions of pay allocation, particularly pay equality. For instance, a published study by Mas¹⁰ and unpublished studies by Baker et al.¹¹, Bennedsen et al.¹², and Cullen and Pakzad-Hurson¹³ all point to pay transparency prompting organizations to make pay more equal, including more equal by gender and rank. Cullen and Pakzad-Hurson develop a formal model predicting pay compression following transparency as a result of bargaining concerns. They show corroborating evidence from the

private sector of the US economy using a staggered rollout of policies facilitating communication about pay between co-workers. In a working paper that explores Canadian academics' salaries, Baker et al. leverage the staggered introduction of pay transparency laws across Canadian provinces (and partly within institutions) to show that pay transparency is associated with more equal pay by gender, essentially a narrowing of a gender pay gap that is adjusted for rank but not individual performance. Gartenberg and Wulf¹⁴ find that pay transparency is associated with a diminished relationship between division manager pay and division performance within geographically dispersed firms. While these studies explore pay transparency's influence on specific elements of pay in organizations, decisions about pay equity, pay equality and pay for performance are highly interrelated. Our aim in this paper is to examine pay transparency's systemic influence on an organization's pay practices, specifically pay equity (including the performance-conditioned gender pay gap), pay equality and pay for performance.

To explore pay transparency's broad influence on pay requires access to rather unique data. Ideally, we would observe a large panel of individual employment data that includes both performance and salary histories surrounding exogenous shocks to pay transparency. The US academic context provides an appealing setting to assemble such data. First, many key individual productivity outcomes are observable and measurable, enabling relatively reliable estimates of both discriminatory and non-discriminatory wage differentials, including the performance-conditioned gender pay gap, as well as estimates of pay for performance. Second, through the Freedom of Information Act and state-specific Sunshine laws, salary data for public university employees have become available in most states, albeit archived in widely disparate data repositories and with varying ease and cost of access. Finally, in the past decade, a wave of transparency events in the form of published websites dramatically eased university employees' access to peer salary data. These websites appeared in a staggered fashion essentially state by state, but each well after the imposition of the Freedom of Information Act and state-specific Sunshine laws. Our data combine detailed information about the individual academic performance of close to 100,000 US academics (that is, their publications, awards, grants,

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Table 1 | The effect of pay transparency on the gender wage gap

DV: ln(wage)	Model 1	Model 2	Model 3	Model 4
Treatment	0.006, $t_{138}=0.60$, (0.011), 0.55, (−0.01, 0.03)	−0.009, $t_{138}=1.04$, (0.009), 0.30, (−0.03, 0.01)	−0.009, $t_{85}=1.10$, (0.008), 0.28, (−0.02, 0.01)	−0.015, $t_{85}=1.82$, (0.008), 0.07, (−0.03, 0.00)
Female	−0.112, $t_{138}=12.41$, (0.009), <0.001, (−0.13, −0.09)	Absorbed	−0.062, $t_{85}=11.93$, (0.005), <0.001, (−0.07, −0.05)	Absorbed
Treatment × female	0.059, $t_{138}=7.22$, (0.008), <0.001, (0.04, 0.07)	0.031, $t_{138}=7.44$, (0.004), <0.001, (0.02, 0.04)	0.020, $t_{85}=3.99$, (0.005), <0.001, (0.01, 0.03)	0.021, $t_{85}=5.76$, (0.004), <0.001, (0.01, 0.03)
Associate professor			0.121, $t_{85}=14.95$, (0.008), <0.001, (0.10, 0.14)	0.064, $t_{85}=8.79$, (0.007), <0.001, (0.05, 0.08)
Full professor			0.391, $t_{85}=33.25$, (0.012), <0.001, (0.37, 0.41)	0.173, $t_{85}=12.77$, (0.014), <0.001, (0.15, 0.20)
Productivity controls	Yes	Yes	Yes	Yes
Individual fixed effects	No	Yes	No	Yes
Academic field (11 categories) fixed effects	Yes	Absorbed	Yes	Absorbed
Institution fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Observations	306,404	300,853	195,976	194,077

The table presents OLS regression estimates explaining (ln) salaries. DV, dependent variable. The results are shown as follows: β , t_{df} , (s.e.), *P* value, (95% confidence interval (CI)). Productivity controls include academic tenure (ln), number of academic articles, number of books, number of awards, number of grants and number of patents. In models 5 and 6, the omitted category is assistant professor. Standard errors are clustered at the level of institution. See Supplementary Table 3.2 for additional specifications.

books and patents) with their demographic characteristics (gender (as estimated dichotomously from first names), rank, tenure and discipline) and their salary histories between 1997 and 2017. We then exploit staggered shocks to the accessibility of information on wages in the public university systems in the United States to explore how pay transparency changes pay equity and pay equality, as well as the performance basis of pay—specifically, how the links between pay and observable performance measures change both within the broader population and within individual academic departments and institutions.

Our results suggest that pay transparency has an economically sizeable effect in reducing pay inequity, significantly reducing the performance-conditioned gender pay gap, as well as more broadly improving the precision with which pay is linked to observable performance metrics and promotion. Overall pay allocation becomes more fair, more equitable and less discriminatory, at least on the performance dimensions we can measure. At the same time, our results suggest that pay transparency has a significant and economically sizeable effect in increasing the equality of pay, reducing by nearly 20% the level of pay variance within departments and institutions. Overall, pay also simply becomes more equal.

One way an organization composed of individuals with heterogeneous abilities can generate both more equal and more equitable pay is to have pay both more precisely and more weakly linked to individual performance. We find evidence of exactly this: pay transparency leads to significant and economically sizeable reductions in the performance basis of pay. Not only is pay more consistently and equitably linked to performance, but the financial rewards linked to observable performance metrics as well as rank advancement substantially decline after wages become more transparent.

In aggregate, our results confirm that pay transparency has the consequences that many policy advocates claim. It prompts organizations to reduce inequity and inequality in pay allocation. At the same time, pay transparency has consequences less frequently discussed. Pay transparency prompts those allocating pay to weaken the link between observable performance metrics and pay. We view our results as providing an empirical test of the causal effect of pay transparency on three fundamental attributes of pay—pay equity, pay equality and the performance basis of pay in organizations—thereby generating a framework for policymakers and practitioners alike to evaluate and debate the arguably complex consequences of pay transparency.

Results

Wage transparency and pay equity. We begin by analysing the impact of pay transparency on pay equity—the fairness and consistency with which an institution or department allocates pay to individuals. Empirically, we operationalize equitable pay as a ‘market wage’ or pay that is predicted, in any given year, by observable productivity outcomes (published articles, books, grant funding, patents, awards and academic rank), academic experience, and institutional and academic field affiliation. Unfair pay, at an individual level, falls above or below this estimated market wage, while discriminatory pay, such as gender-based discrimination, is evident from a category’s systematic deviation from this predicted fair wage. We acknowledge that this operationalization may mask important inequities that are driven by hiring and promotion processes, allocation of tasks or discrimination that plays out in the generation of productive outcomes such as publications, awards or grants^{15–18}. We recognize that such discrimination will be hidden in our approach and unobservable in assessing who is unfairly overpaid or underpaid. Individuals may also have differing beliefs about whether the

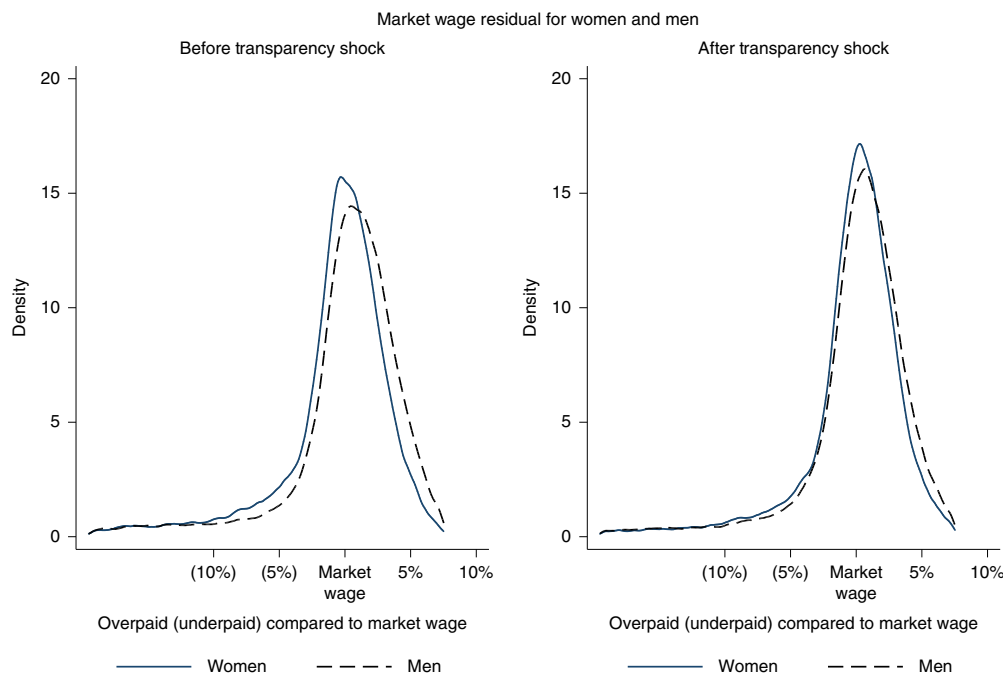


Fig. 1 | Equity in organizations: distribution of market wage residuals by gender and transparency shocks. Kernel density estimates of (ln) wage regression residuals by gender and transparency shocks are shown. Controls include academic tenure (ln), number of academic articles, number of published books, number of awards, number of grants and number of patents, as well as institution, academic domain and year fixed effects. To allow comparison, all models are run jointly for men and women. The residuals are trimmed at 1% and 99%. Two-sample combined Kolmogorov-Smirnov tests for equality of distribution functions: 0.124, $P < 0.001$ (left); 0.067, $P < 0.001$ (right).

academic market is fairly weighting specific variables as measured in our models. Our interest, though, is in whether pay transparency reshapes the consistency with which pay is allocated to these observable measures.

One of the key dimensions of inequity in the allocation of wages concerns discriminatory practices based on gender. Indeed, the proponents and regulators in favour of greater wage transparency often claim that such policies are likely to be an efficient tool in detecting and forcing organizations to eliminate this precise form of discrimination^{8,19}. In the results that follow, we first present evidence of pay transparency's influence on what we call the conditional gender pay gap—the gender pay gap after controlling for rank and performance—and then move to examining pay transparency's influence on the equitability of pay allocation more broadly.

Our data suggest that both the unconditional gender wage gap (not controlling for performance or rank measures) and the conditional gender wage gap have been decreasing steadily over the past two decades in our sample of academics, though both gaps continue to be sizeable (Extended Data Fig. 1 and Supplementary Table 3.1). Controlling for institutional affiliation, academic discipline, tenure and productivity (article and book publications, grants, patents and awards), an average female academic in our sample was paid 7.7% less than her male counterpart in 2010. While there is important heterogeneity across states and academic disciplines (Extended Data Table 1), our data indicate that women continue to be underpaid compared with men across the eight states that we study and across academic disciplines. By 2017, the average conditional gender pay gap had narrowed but remained substantial at 2.6%.

This decreasing level of inequity can be partially explained by shocks to the accessibility of information on salaries. To explore how these transparency shocks differentially affected subsequent wages of men and women academics in our sample, we specify the following econometric model explaining individual i 's wages in year t ($Y_{i,t}$):

$$\ln(Y_{i,t}) = F_i + \sum_k \beta_k \text{Treatment}_{i,t}^k + \sum_k \gamma_k \text{Treatment}_{i,t}^k \times F_i + \text{Controls}_{i,t} + e_{i,t} \quad (1)$$

where k corresponds to the number of lags (set in the dynamic model to three short-term lags and one long-term lag) and leads (the year following the shock, three short-term leads and one long-term lead) of the treatment event. β_k represents the marginal effects of year from (to) transparency shock on (ln) wages. The model further includes time-varying productivity controls and year, institution and individual fixed effects. As institutions don't change states over time, we omit state fixed effects from our models; when individual fixed effects are omitted, we additionally include academic domain fixed effects. F_i is an indicator variable taking a value of 1 for female academics and 0 otherwise; $e_{i,t}$ is the error term. All models reported in the paper are ordinary least squares (OLS) regressions with correction for multiway fixed effects implemented in the `reghdfe` Stata (version 16) package²⁰.

The coefficients of initial interest are γ_k , representing the marginal effect of year from (to) transparency shock on the wages of female academics compared with a baseline (male academics). In Table 1, we report the full multivariate results based on the static or canonical specification (omitting individual lags and collapsing all leads to one treatment dummy), allowing us to better quantify the size of these effects. In our basic models, we cluster errors at the level of the institution. Our results are robust to clustering at the state-year and state level (see Supplementary Section 6 for more details on the calculation of standard errors with few clusters). For comparison purposes, we report additional results in Supplementary Table 3.2. In models including individual fixed effects, the dummy variables for female and academic domain fixed effects are absorbed. Depending on the model, we estimate that the transparency shocks led to the conditional gender pay gap closing by a range of 2 to 5.9 percentage points. The magnitude of these effects is comparable to

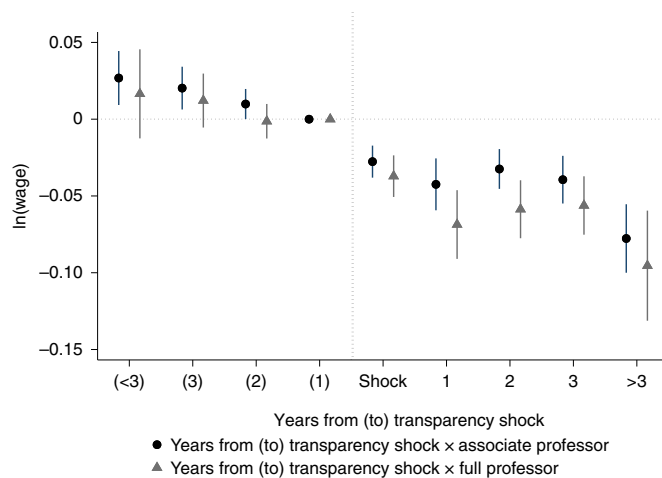


Fig. 2 | The effect of wage transparency on salary adjustments associated with promotions. Regression coefficients from a dynamic difference-in-differences OLS regression model explaining $\ln(\text{wage})$ are shown. The reference category is one year prior to transparency shock. The plotted coefficients are for years from (to) transparency shock interacted with rank, with 95% CIs. Vertical dotted line separates pre- and post-treatment estimates and horizontal dotted line marks no change in $\ln(\text{wage})$. The models are run jointly for all ranks. The errors are clustered on institution. The controls include year, institution and individual fixed effects. The regression results used to generate this plot are reported in Supplementary Table 5.4.

results reported by Baker et al.¹¹, who estimate a 2-percentage-point reduction in the gender pay gap resulting from changes to transparency laws in Canada.

In Extended Data Fig. 2, we plot coefficients from a dynamic specification including individual lags and leads; the full regression results are reported in Supplementary Table 3.3. A visual inspection indicates a noticeable increase in the relative wages of female academics compared with male academics in the years following the transparency shocks. Although we observe no pre-trend in the three years preceding the transparency shock, the long-term lag coefficient is negative. While the long-term lag coefficient is less readily comparable with other estimates because of the composition effect, a pre-trend could be of concern. To address this concern, we conduct five additional sets of analyses. First, we restrict our data to the years 2004–2011 or 2004–2013 inclusive. This ensures that at least one state (Pennsylvania in the sample restricted to 2013 and Pennsylvania and Florida in the sample restricted to 2011) is never treated within our observation period. The results of the model restricting the data to 2013 are presented in Extended Data Fig. 3a and show a pattern that is consistent with our earlier estimates. Due to a shorter right-side time series of our data, we report only the period following the shock, one short-term lead and one long-term lead estimate in this specification. Second, we employ a ‘stacked’ difference-in-differences approach²¹. For each of our first three cohorts of treatment (Cohort 1: West Virginia, shock in 2007; Cohort 2: New York and California, shock in 2008; Cohort 3: Texas, shock in 2009), we create three separate datasets that include only cohort-specific states and, as a control group, states that have not been treated in the two years following the shock. For example, Cohort 2 includes New York and California as treated states and Pennsylvania and Florida as the control group, with academics from all other states dropped. We then stack these three datasets, centring on the timing of the treatment, including three lags, the year following the treatment and two leads of the transparency event. We further saturate the fixed effects with cohort dummies. We present

the results of the dynamic model fit to these data in Extended Data Fig. 3b and show that the stacked specification does not suffer from a pre-trend in the three years preceding the shock. Third, to directly tackle the possible pre-trend concern, we rely on an instrumental variable estimation strategy²². We implement a two-stage least squares estimator wherein a covariate—in our setting ideally showing similar trends to the gender pay gap pre-transparency shock but unaffected by the treatment—is included in the regression and instrumented with a one-year lead of the transparency shock. In the reported specifications, we use the state-year measure of unconditional gender wage gap in the private sector. We present the results of this model in Extended Data Fig. 3c. Details of the additional data used for this analysis are discussed in Supplementary Section 3. Fourth, to gain stronger insight into a potential pre-trend, we drop from our analyses the three earliest-treated states (West Virginia, California and New York) and hence restrict our data to states for which we have a longer panel of pre-treatment data. Extended Data Fig. 3d,e reports the results from a dynamic model fit to these data that partially alleviates concerns that our estimated results are driven by anticipatory wage adjustments. Finally, in Extended Data Fig. 3e, we report the results from a fully unconstrained model (including all possible leads and lags of transparency shocks) along with fitted trend lines. These supplemental analyses, more fully documented in Supplementary Tables 3.3–3.8, reduce pre-trend concerns and strengthen support for a conclusion that pay transparency reduces the performance-conditioned gender pay gap.

Another approach to exploring the influence of pay transparency on the gender pay gap is to simply visualize how the precision with which pay is predicted by observables changes after pay transparency for men and women. To generate this visualization, we plot how wage transparency affects the distribution of residuals from our market wage regressions. For each academic, we predict yearly ‘fair’ market wages on the basis of their institutional affiliation, academic domain affiliation, academic tenure and productivity outcomes (see also ‘Mechanisms’ for a more detailed discussion of the market wage estimation). Figure 1 shows kernel density plots of residuals for female and male academics before and after transparency shocks. Actual, observed wages below this predicted wage are considered inequitable underpayment, while observed wages above this predicted wage are considered inequitable overpayment. The pattern in the data is quite clear. Prior to transparency shocks, women are substantially more likely to be underpaid (compared with the estimated market wage) than are men. This can be seen from a larger mass of the distribution of residuals to the left of the estimated market wage. Women are also less likely to be overpaid (compared with the estimated market wage). Indeed, the overall distribution of residuals from the market wage is shifted to the left and narrower for women, compared with men. Following the pay transparency shocks, the two distributional plots become substantially more aligned. Although women are still more likely to be underpaid and less likely to be overpaid than men, these differences become smaller after the transparency shocks. These results echo our earlier estimates, indicating that although increased access to wage information had a positive causal effect on closing the gender pay gap, some unexplained differences continue to be present in our data.

The gender pay gap is, of course, but one manifestation of possible inequitable wage practices in organizations. Full exploration of all discriminatory factors is well beyond the scope of this paper. However, we also investigate the extent to which pay transparency leads to an increase in the overall equity or consistency with which pay is associated with productivity and rank. To perform these analyses, we measure the variance in residuals from ‘fair’ market wage regressions within academic fields by institution. This measure takes on smaller values in departments where most faculty are equitably paid and where faculty are on average consistently underpaid or overpaid relative to the broader market, but it takes on larger

Table 2 | The effect of pay transparency on institution-academic field variance in market wage residuals and variance in (ln) wages

	Model 1	Model 2
DV	Variance in market wage residuals	Variance in (ln) wages
Within	Institution-academic field (11 categories)	
Treatment	−0.034, $t_{130} = 2.27$, (0.015), 0.025, (−0.06, −0.01)	−0.065, $t_{136} = 4.49$, (0.015), <0.001, (−0.09, −0.04)
Institution-academic field (11) fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Observations	9,015	12,892

The table presents OLS regression estimates explaining institution-academic field variance in residuals from market wage regressions (model 1) and variance in wages (model 2). The results are shown as follows: β , t_{stat} , (s.e.), P value, (95% CI). Market wage regression controls include academic tenure (ln), number of academic articles, number of published books, number of awards, number of grants and number of patents, as well as institution, academic domain and year fixed effects. The residuals are trimmed at 1% and 99%. Standard errors are clustered at the level of institution. See Supplementary Tables 4.1 and 4.2 for additional results.

values in departments where the determinants of pay allocation are uneven and vary substantially across members. We specify the following econometric model explaining variance in fair wage residuals in year t ($\text{Var}R_{j,t}$) by academic field-institution (j), using either the 11 or 25 academic field categories:

$$\text{Var}R_{j,t} = \sum_k \beta_k \text{Treatment}_{j,t}^k + \text{Controls}_{j,t} + e_{j,t} \quad (2)$$

where the controls include mean levels and variance in all productivity outcomes by academic field, with j and t fixed effects, and k defined analogously to equation (1). Model 1 in Table 2 presents the results of a static difference-in-differences specification, using the 11 academic fields specification. We find that wage transparency decreased the variance in residuals by 11%, on average. Finally, in Extended Data Fig. 4 (left), we present results of the dynamic specification that reveal patterns consistent with the results previously discussed. These analyses indicate that, if anything, inequity in pay allocation had been rising in the years leading to transparency shocks—a trend that was reversed around the time of treatment. Taken together, our results indicate that increased access to wage information resulted in decreasing various forms of wage inequity within academia.

Wage transparency and pay equality. Unlike equity, equality is an absolute construct^{3,23}, and when applied in its strongest form, full equality would imply no variance in wages across all individuals, independent of either performance or rank. Although we could theoretically imagine compression taking place across all sectors of an economy or across all institutions and disciplines within academia, we explore whether wage transparency results in greater equality defined as lesser dispersion in wages among department peers, independent of their performance.

Model 2 in Table 2 presents the first set of analyses of the impact of pay transparency on equality of pay. We report regression results of the static difference-in-differences specification explaining changes to institution-department wage variance on the basis of the calculation of variance across different reference categories defined by institutional and academic field affiliation, as in the model below:

$$\text{Var}Y_{j,t} = \sum_k \beta_k \text{Treatment}_{j,t}^k + \text{Controls}_{j,t} + e_{j,t} \quad (3)$$

where $\text{Var}Y_{j,t}$ corresponds to the field-institution-level variance in (ln) wages, and all other parameters are specified identically to those in models estimating equation (2). Our results are consistent across models and imply that wage transparency had a strong effect in prompting equality through reduced wage dispersion. The long-term impact is economically sizeable. We find that wage transparency decreased pay dispersion by nearly 20%.

We also analyse the results of dynamic specifications. Our results (reported in Extended Data Fig. 4, right) corroborate the role of transparency shocks in decreasing the average wage variance of academics in our data and the lack of explanatory pre-trends. To further explore the mechanism behind pay transparency's impact on wage equality, we again turn to distributional plots. This time, however, we focus on wage equality and hence on wage residuals—before and after transparency shocks—from regression models predicting wages on the basis of institutional and department affiliations, while controlling for temporal variation using calendar-year fixed effects. Consistent with our earlier results, the evidence emerging from these analyses indicates that pay becomes more equal or simply more compressed following transparency shocks. This narrowing of the wage distribution can be attributed to reducing the density of both positive and negative wage residuals (see Extended Data Fig. 5 for kernel density plots).

Mechanisms. The prior two sections provide evidence that pay transparency prompts departments and institutions to elevate both pay equity and pay equality, but our results to this point offer little visibility into precisely how this has occurred. In this section, we explore three possible mechanisms. First, pay transparency may simply heighten pressure to weaken the relationship between pay and performance. Second, pay transparency may prompt institutions to focus on adjusting the pay of those most underpaid or overpaid, measured on the basis of either equality or equity. Finally, pay transparency may prompt mobility (employee entry and exit) that mechanically elevates pay equity and/or equality. For instance, those who discover that they are unfairly underpaid through pay transparency may depart for other institutions, leaving those more fairly paid to remain. We briefly explore evidence for each of these mechanisms.

To explore the first mechanism—a weakening pay-for-performance relationship—we estimate a series of models predicting the effect of transparency shocks on the relationship between both productivity outcomes and academic rank and salary. We thus specify the following general model explaining (ln) wages:

$$\ln(Y_{i,t}) = \text{Treatment}_{i,t} + \sum_l \beta_l P_{i,t}^l + \sum_l \sigma_l P_{i,t}^l \times \text{Treatment}_{i,t} + \text{Controls}_{i,t} + e_{i,t} \quad (4)$$

where $P_{i,t}^l$ corresponds to individual i 's productivity on metric l in year t , and Treatment is a dummy variable equal to one for all years subsequent to a transparency shock. Alternatively, we substitute productivity outcomes with academic rank (associate and full professor, with assistant professor being the reference category). Controls include individual, institution and year fixed effects. The coefficients β_l indicate the average strength of the link between pay and performance, while the coefficients σ_l indicate how a transparency shock affects this average link. Negative σ_l coefficients indicate weakening of the marginal returns, compared with the baseline pre-transparency levels, while positive σ_l coefficients signify increased weight on performance in determining wages.

The results indicate that, following transparency shocks, institutions in our data began to rely less strongly on pay for performance and that salary differences across academic ranks also fell substantially. Although all the productivity measures that we observe had, and largely continued to have, a positive effect on salaries (as do

Table 3 | Predicted marginal effects of star levels of performance and rank advancements on wages before and after transparency shocks

	Population		Humanities		Physical and mathematical sciences		Biological and biomedical sciences		Social and behavioural sciences		Engineering	
Before/after transparency	Before	After	Before	After	Before	After	Before	After	Before	After	Before	After
Academic articles	10.8%	5.6%	2.8%	2.3%	12.9%	8.6%	21.2%	8.3%	10.6%	6.0%	18.0%	9.9%
Patents	0	−0.2%	0	0	2.6%	0.3%	0	0	0	0	4.2%	4.2%
Books	3.3%	1.6%	8.5%	4.9%	0	0	0	0	4.8%	1.6%	0	0
Grants	8.1%	5.8%	0	0	14.7%	10.6%	10.8%	10.8%	6.3%	6.3%	7.4%	7.4%
Awards	0	0	0	0	0	0	0	0	0	0	0	0
Promotion to associate (compared with assistant professor)	14.5%	8.7%	12.7%	7.9%	11.4%	4.6%	14.4%	7.3%	14.3%	9.7%	10.3%	4.6%
Promotion to full (compared with assistant professor)	31.9%	25.2%	27.8%	24.1%	22.7%	17.9%	30.6%	23.5%	30.1%	25.3%	18.6%	14.4%

The table presents the predicted marginal effects of performance outcomes on salaries. Star performance levels are calculated as the 95th percentile of the performance distribution across all years, separately for each domain when relevant, and within regression samples. When the 95% CIs of the estimated coefficients include zero, we report the marginal effect to be nil. The regression results used to generate this table are reported in Supplementary Table 5.3a,b.

promotions), the strength of these relationships weakens substantially following transparency (with awards being the sole exception). We report the full results of these regression models in Supplementary Tables 5.1 and 5.3a,b. In Fig. 2, we report the results of the dynamic model tracing the evolution of marginal returns to advancements across ranks. We observe an economically sizeable, discrete drop in the pattern of marginal returns to rank advancements at the time of transparency shocks, compared with the baseline levels.

In Table 3, we summarize the economic magnitude of these changes for a star academic in our sample (in terms of performance outcomes) and document heterogeneity across academic fields. The interpretation of these results is as follows. Controlling for academic tenure, field and institution, pre-transparency, an average academic with star levels of performance (that is, in the 95th percentile) across all metrics could expect to see a 22.2% greater salary than an academic with no output (observable to us). However, after pay transparency shock, we observe a large (42%) drop in the sensitivity of pay to this composite performance score. The results are similar if we compare the average premium in salaries that accompany rank. In our data, the marginal pre-transparency pay increases associated with promotion to associate and full professors (compared with assistant professors) were +15% and +32%, respectively. After pay transparency, these premiums are still substantial but fell to +9% and +25%, respectively.

Our results also reveal a consistent pattern across academic disciplines, although, as expected, the sensitivity of pay to different performance outcomes varies by field. For example, of the five disciplines, humanities places the greatest emphasis (in terms of compensation) on book publications, whereas biological and biomedical sciences place the greatest emphasis on publications in peer-reviewed journals. At the same time, across all academic disciplines, the sensitivity of pay to performance decreased following transparency shocks. Similarly, across disciplines, the financial rewards associated with promotion across academic ranks fell markedly in response to pay transparency.

To explore a second mechanism, we examine more precisely whose pay is reshaped after pay transparency. Do institutions decrease the pay of those inequitably overpaid or increase the pay of those underpaid? To explore these adjustment patterns, we first predict the ‘fair’ market wage in year t for individual i , $\ln(\widehat{Y_{i,t}})$, on the basis of their productivity (number of published academic articles, number of published books, number of awards, number of grants

and number of patents), academic tenure, institution and academic field. Additionally, in our predictions, we account for heterogeneity in the sensitivity of pay to different performance measures across academic domains.

Armed with these estimates, we test the extent to which deviations in observed salaries from predicted levels are informative about the changes in wages in the following year. For ease of interpretation and calculation of the economic magnitude of our effects, we generate binary measures of upward and downward inequity, coding individuals as underpaid if their salary is below mean levels of negative residual and as overpaid if their salary is above mean levels of positive residual. The reference category (that is, ‘fairly paid’) includes those whose salaries are between these two levels. In the supplementary analyses, reported in Supplementary Table 5.5, we also rely on a continuous residual measure and use a spline model, separately introducing positive (overpayment) and the absolute value of negative (underpayment) residuals. We therefore estimate several variants of the following model predicting wage adjustments:

$$\frac{Y_{i,t+1} - Y_{i,t}}{Y_{i,t}} \times 100 = \alpha_1 + \beta_1 \text{Treatment}_{i,t} + \beta_2 \text{Upaid}_{i,t} + \beta_3 \text{Opaid}_{i,t} + \gamma_1 \text{Treatment}_{i,t} \times \text{Upaid}_{i,t} + \gamma_2 \text{Treatment}_{i,t} \times \text{Opaid}_{i,t} + \text{Controls}_{i,t} + e_{j,t} \quad (5)$$

where Upaid and Opaid correspond to the measures of underpayment and overpayment, compared with the fair market wage. Controls include academic field, institution and year fixed effects. In some models, we also include a control for the absolute level of wages along with its interaction with the treatment dummy. The main parameters of interest are β_2 , β_3 , γ_1 and γ_2 , indicating, respectively, the sensitivity of pay increases to negative deviations from market wage (underpayment), positive deviation from market wage (overpayment) and a change in these sensitivities following transparency shocks. The dependent variable, wage change, is multiplied by 100 and thus expressed in percentages.

Our results, reported in Table 4, suggest that institutions generally grant larger salary increases to those we estimate as previously underpaid and smaller increases to those we estimate as previously overpaid. Academics in our sample received an average inflation-adjusted yearly salary increase of 2.8% (model 2). Our results indicate that pre-transparency, those who were underpaid

Table 4 | The effect of market wage and pay transparency on yearly wage changes

Underpaid and overpaid (binary specification)			
DV: wage change (%)	Model 1	Model 2	Model 3
Treatment		−0.157, $t_{129} = 0.38$, (0.413), 0.70, (−0.98, 0.66)	−0.347, $t_{129} = 0.87$, (0.385), 0.385, (−1.13, 0.44)
Underpaid	1.372, $t_{129} = 8.83$, (0.155), <0.001, (1.06, 1.68)	1.028, $t_{129} = 4.65$, (0.221), <0.001, (0.59, 1.47)	0.766, $t_{129} = 4.43$, (0.173), <0.001, (0.42, 1.11)
Underpaid × treatment		0.531, $t_{129} = 2.75$, (0.193), 0.007, (0.15, 0.91)	0.516, $t_{129} = 3.17$, (0.163), 0.002, (0.19, 0.84)
Overpaid	−1.568, $t_{129} = 15.16$, (0.103), <0.001, (−1.77, −1.36)	−1.582, $t_{129} = 9.82$, (0.161), <0.001, (−1.90, −1.26)	−1.452, $t_{129} = 10.49$, (0.138), <0.001, (−1.73, −1.18)
Overpaid × treatment		0.023, $t_{129} = 0.17$, (0.135), 0.864, (−0.24, 0.29)	0.161, $t_{129} = 1.29$, (0.125), 0.200, (−0.09, 0.41)
Low salary			0.528, $t_{129} = 2.20$, (0.240), 0.030, (0.05, 1.00)
Low salary × treatment			0.282, $t_{129} = 1.28$, (0.219), 0.202, (−0.15, 0.72)
Academic field fixed effects	Yes	Yes	Yes
Institution fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Observations	259,624	259,624	259,624

DV = $\frac{Wage_{it,t+1} - Wage_{it,t}}{Wage_{it,t+1}} \times 100$. The results are shown as follows: β , $t_{d.f.}$, (s.e.), P value, (95% CI). Underpaid is equal to 1 if individual i 's residual from regression predicting market wage in year t is negative and smaller than the average residual from the same regression for all individuals in a given year, the same institution and the same academic domain. Overpaid is equal to 1 if individual i 's residual from regression predicting market wage in year t is positive and greater than the average residual from the same regression for all individuals in a given year, within the same institution and the same academic field. Low salary is equal to 1 if individual i 's salary is below average compared with year–institution–domain peers and 0 otherwise. Standard errors are clustered at the level of institution. See Supplementary Table 5.5 for additional results including continuous specifications of the overpaid and underpaid variables.

compared with their institution–department peers received yearly increases of 3.8% on average, while those who were significantly overpaid relative to their peers received salary increases of 1.2%, on average. Post-transparency, we observe that the average wage increases for those underpaid rise to 4.3%, while increases for those overpaid remain unchanged.

We also test whether these results are simply driven by differences in the absolute levels of pay (which are correlated with distance from market wage). Over and above simply controlling for absolute levels of salaries, in model 3, we add a dummy variable and its interaction with treatment for low absolute levels of salaries, defined as salaries that are inferior to the year–institution–domain average pay. High salary is the reference category. The results are consistent across models and suggest a narrative that pay transparency heightens attention to both pay equity and pay inequality and that after transparency institutions seek to remedy inequity and inequality by actively increasing the pay levels of those most underpaid as measured by both equity and equality.

We explore one final mechanism that may explain our pay transparency results on equity and equality. All our results presented above span the full population of academics, including those switching institutions and those leaving or joining our sample during the observation period. It is conceivable that shifts in the

sample population resulting from employee mobility prompted by pay transparency may explain these increases in pay equity and pay equality. For instance, those receiving low pay, whether fairly or unfairly, may depart in response to pay transparency, leaving those more highly and equitably paid to remain. The net effect may be that those who remain are both more equally and more equitably paid. A full exploration of this mobility mechanism is beyond the scope of this paper. However, to validate that this mobility pattern is not driving our results, we drop from our sample all academics who have (1) changed institutions within our observation window but stayed in our sample, (2) left our observation sample between transparency shock and the end of the observation window, or (3) joined our sample after the transparency shocks. We thus restrict our analysis to the ‘non-mobile’ workforce. We rerun all our models using this restricted sample and find consistent results throughout. Supplementary Table 3.7 reports results mirroring those in Table 1 but using the restricted sample. Our results continue to show that transparency shocks decrease the gender pay gap across specifications, although our estimates are generally slightly smaller in economic magnitude, suggesting that mobility patterns may indeed play some role in explaining the influence of pay transparency on the declining gender pay gap. We similarly test whether our first mechanism results—a weakening pay-for-performance relationship

summarized in Table 3 and Supplementary Table 5.1—hold with this restricted sample. Our results (Supplementary Table 5.2) are indeed robust to using this restricted sample, supporting the claim that wage transparency has led to a significant change in organizational pay practices that have fuelled shifts in the equality and equity of pay allocation.

In Supplementary Section 6 and the Methods, we discuss an additional series of robustness tests with respect to inferring the statistical significance of our results, sample construction and the robustness of our results to omitting data from larger states. Importantly, neither the exclusion of Texas nor the exclusion of California, the two most populous states that also generate the most data, significantly changes our results.

Discussion

Our results suggest that in the context of academia in the United States, pay transparency has a systemic and sizable effect on the structure of pay. While earlier work has particularly shown pay transparency to prompt more equal pay^{10,11,13}, we provide an empirical test of the broader causal effects of pay transparency on pay allocation, including pay equity—the fairness or consistency with which pay is matched to performance and rank, as well as the overall performance basis of pay. Pay transparency seems to pressure those who assign pay to more aggressively remedy inequities in the allocation of pay, granting larger pay increases to those who, on the basis of their performance and rank, are unfairly underpaid. The performance- and promotion-conditioned gender pay gap is significantly reduced, but more generally, pay becomes more precisely predicted by observable performance measures. Pay transparency also seems to pressure those responsible for allocating wages to simply make pay more equal, independent of performance. Pay becomes more compressed, and department and institution affiliation predict a larger portion of pay variance. Finally, consistent with pay becoming more equal, in response to pay transparency, academic departments and universities significantly weaken the link between pay and a range of observable performance metrics, including publications, books and grants. In addition, the marginal returns to advancements in academic rank become significantly smaller. In summary, pay becomes more equal and more precisely assigned to observable metrics, but also less performance- and promotion-based.

Prior work on pay transparency has explored a variety of individual psychological and behavioural responses to pay transparency relating to happiness, satisfaction or desire to exit^{24–26}—responses that will vary on the basis of what individuals discover about where their individual level of pay ranks relative to others. Underlying these results is the reality that humans socially compare, and when they perceive pay inequity or pay inequality, they experience emotions of injustice or envy that may reduce job satisfaction or well-being^{27–29}. This in turn may prompt behaviours costly to employers, including turnover, reduced effort and social cohesion, or simply politicking for change in pay allocations^{30–33}. Enhancing the visibility of pay enhances the visibility of inequity or inequality, heightens these emotions of envy and injustice, and elevates these attendant costly responses, which in turn elevates pressure on employers to seek greater equity and equality.

Our results focus on pay transparency's influence on organization-level responses. But, of course, individual and organization-level responses to pay transparency are interrelated, and our understanding of pay transparency would greatly benefit from future work exploring whether and how these individual effects function as mechanisms that deliver these organization-level responses to pay systems. For instance, prior work shows that both men and women may consider comparatively lower female wages as fair³⁴. Future work could fruitfully study the extent to which transparency affects (or does not affect) such justice perceptions.

Card and colleagues²⁵ results, also in an academic setting, suggest that pay transparency prompts those receiving below-median pay to quickly express lower job satisfaction and higher turnover intentions, but actual turnover measured after several years shows a much noisier relationship with relative salary levels. As our study shows, academic institutions are dynamically adjusting pay in response to pay transparency and doing so in patterns that render pay more equal and equitable. Those initially expressing turnover intentions in response to discovering inequity or inequality are precisely those likely to receive larger pay increases, which may quell their initial motivation to depart. Pay transparency may also have important implications for performance and productivity. Breza et al. find that when wages and productivity are visible and inequitably allocated, productivity and cooperation decline³⁵. Perez-Truglia³⁵ finds that pay transparency increases the well-being or happiness gap between those receiving high and low pay—a result that may indirectly contribute to diminished productivity and performance. Precisely how pay transparency jointly shapes both individual and organizational outcomes merits deeper exploration.

For policymakers, including managers responsible for decisions regarding pay transparency, our results illuminate what some might consider an important trade-off between increased equity and equality and weakened pay for performance. How policymakers differentially value these pay allocation outcomes will of course weigh heavily in the decision of how transparent to be. Our results may also illuminate why organizations and some policymakers seem to prefer choices to enhance pay transparency that fall short of broadcasting individual levels of pay. Such transparency-increasing practices may reveal the structure by which pay is assigned, such as revealing pay ranges by hierarchical rank or disclosing pay levels for relevant aggregated groups of peers rather than individual wages for the full set of peers. Such practices may pressure organizations to elevate the fairness and consistency of pay while still maintaining pay for performance. The effectiveness of these alternative forms of pay transparency merits further exploration.

Relative to many other work contexts, ours is one in which individual performance is rather observable. While important elements of academic output are team-based, the diverse array of teams on which any given individual works (as well as a general pattern of positive assortative matching) yields a reasonably good assessment of individual contributions, at least relative to many other work settings. The fact that even in this environment, one with reasonably visible and objective measures of performance, pay transparency still generates these strong effects on pay equality suggests that our findings may generalize to other contexts. In settings with less visible performance measures, we might predict even stronger pressure to equalize pay—a prediction that Cullen and Pakzad-Hurson¹³ explore theoretically. Nonetheless, how well our findings generalize to other settings in which pay is made transparent is an important limitation of the study.

Methods

Data. To compile the required data, we first obtained privileged access to a database of individual academic productivity compiled by Academic Analytics, an analytics and consulting service provider to universities and educational institutions. These data were meticulously assembled from publicly available sources for the years 2004–2017 and provide information on individual article publications, books, patents, awards and grants. The data encompass all full-time research faculty across all academic fields employed at 412 PhD-granting institutions in the United States, including all 141 AACSB-accredited universities, all 262 R1 and R2 universities, and most of the 161 D/PU institutions.

We then sought matching salary histories for those employed within the public institutions covered by Academic Analytics. While considerable salary data could conceivably be scraped from the publicly accessible websites that appear in each state, these websites generally provide only historical data that begins after the date of the website's publication. Since we require data both before and after transparency shocks, we solicited access to salary data through official Freedom of Information Law requests to state comptroller offices and individual public universities in all 50 states. As a general pattern, only more populous states were able or willing to

provide the necessary salary data in a digital format (see Supplementary Section 2 for more details). From these requests, we obtained yearly employment and salary information for nearly all public university employees in eight states: California, Connecticut, Florida, New York, Pennsylvania, Texas, Virginia and West Virginia. These responses encompassed data from 139 institutions spanning 1997 through 2017 (inclusive; see Supplementary Table 1.1 for details and exceptions).

To then merge the salary data with the productivity data, we developed a matching algorithm based on names and other overlapping individual information. Our final merged dataset spans 676,055 individual-year observations. This includes 97,839 distinct individuals employed in 139 distinct institutions across 11 or 25 broad academic fields (depending on the level of aggregation). We coded gender dichotomously, on the basis of academics' first names. Our study thus does not allow us to account for the full spectrum of gender identities. The coding procedure was carried out by J. Cox. To identify male and female academics, we used the World-Wide Gender Dictionary (<https://ideas.repec.org/c/wip/eccode/10.html>). We kept only names uniquely identified as male or female in this dictionary. We then manually checked the unmatched names and identified 128 additional names to which we could attribute gender with rather high levels of confidence. In particular, we used data from the US Social Security Administration between 1880 and 2020 to calculate the relative frequency of each name for a specific gender (female or male). All but two additional names showed at least 90% consistency in allocation to a gender. We coded two remaining names: Lauren (83% female frequency) and Lindsay (65% female frequency) as female names. The exclusion of these two names does not materially affect our results. A list of 128 additional names along with gender coding is posted with the main data files and code. Using this method, we identified 52,016 individuals as men and 28,839 as women. For models with gender, academic tenure and productivity controls, our sample size decreases to 44,837 individuals due to missing gender data and the fact that our observation window for productivity outcomes begins in 2004. All models including productivity outcomes span 2004–2017. All models without productivity outcomes span the full range of available data (unless explicitly specified otherwise). In all models, we drop 0.5% of the top and bottom earners to account for outliers, and all salaries are expressed in constant 2016 US dollars. Finally, we were able to unambiguously identify rank from job titles for 46,572 individuals in seven out of eight states (with New York as the exception). Given that data limitations constrain us to use a subset of individuals for models that include productivity outcomes, gender and rank information, as a robustness test we rerun all our analyses presented throughout this paper using the most constrained dataset—one that features no missing data on any of the variables. All our results remain robust to this reduced sample of academics and to the shortened time period. Unless otherwise noted, all statistical tests are two sided. Details on the construction of all variables are provided in Supplementary Section 1. Summary statistics for the academics in our final sample are reported in Supplementary Tables 1.2 and 1.3. Additional data sources include the following: measures of the unconditional gender wage gap in the private sector were constructed using the Current Population Survey, conducted jointly by the US Census Bureau and the Bureau of Labor Statistics; journal impact factors were crawled from Journal Citation Reports; and the ranking of academic institutions was constructed on the basis of Academic Ranking of World Universities data.

To explore the causal effect of wage transparency on the constructs of interest, we took advantage of staggered shocks to the accessibility of wage information about public university employees that occurred within the eight states covered by our data. Over the past 15 years, public access to wage information on government employees has been significantly facilitated by the emergence of searchable datasets developed and launched by newspapers, non-governmental organizations and state agencies. Examples of such databases include the California State Worker Salary Database, launched by the *Sacramento Bee* in 2008, and Florida Has the Right to Know, initiated by Florida's governor Rick Scott in 2011. For each state in our sample, we identified the year in which the first such database was launched. We then considered each individual academic as treated if he or she was employed in one of the institutions of the focal state in any of the years following the launch of the database. In the sample of eight states covered in our database, such shocks to transparency happened in a staggered fashion, between 2007 and 2012. Details on the institutional context and all transparency shocks are provided in Supplementary Section 2 and Supplementary Table 2.1.

Additional robustness tests. Over and above the tables reported in the manuscript and the Supplementary Information, we conducted three additional sets of analyses to ensure the robustness of our results. First, we restricted our data to academics employed in highly research-intensive institutions. Our concern in relying solely on pooled results was that the basis of pay may significantly differ across types of institutions. Each year, we ranked 139 institutions present in our data using the Academic Ranking of World Universities. We then reran all our analyses keeping only the 70 most highly ranked institutions and found robust results. Second, we reran the models reported in Table 1 including state-specific time trends and one- and three-year anticipatory effects³⁶. Third, because of particularities of the payment scheme to health professionals (fee-for-service), we reran all our analyses dropping this academic domain from our sample and found robust results.

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

All data that support the findings of this study have been deposited in the Open Inter-university Consortium for Political and Social Research Repository under project number 155541 and are available at <https://doi.org/10.3886/E155541V1>. The names of individual academics and institutions have been blinded and are represented in the data with author-generated unique identifiers.

Code availability

The statistical code generating all results reported in the manuscript has been deposited in the Open Inter-university Consortium for Political and Social Research Repository under project number 155541 and is available at <https://doi.org/10.3886/E155541V1>.

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Author contributions

T.O. and T.Z. jointly conceived the project and supervised the data collection. T.O. conducted the data analyses with input from T.Z. T.O. and T.Z. jointly drafted the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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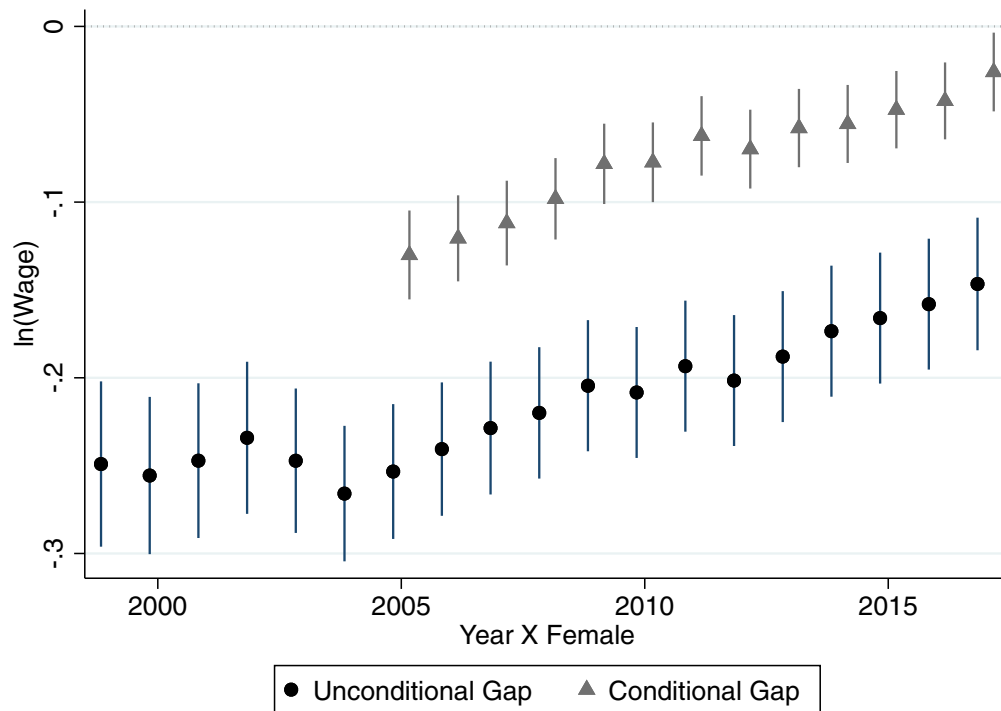
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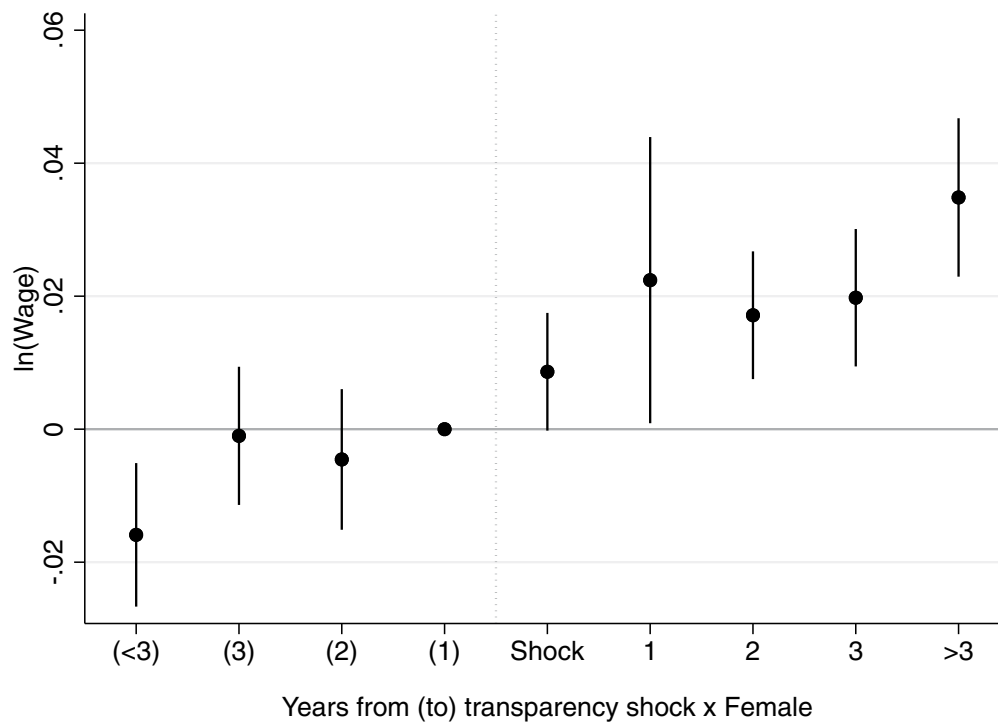
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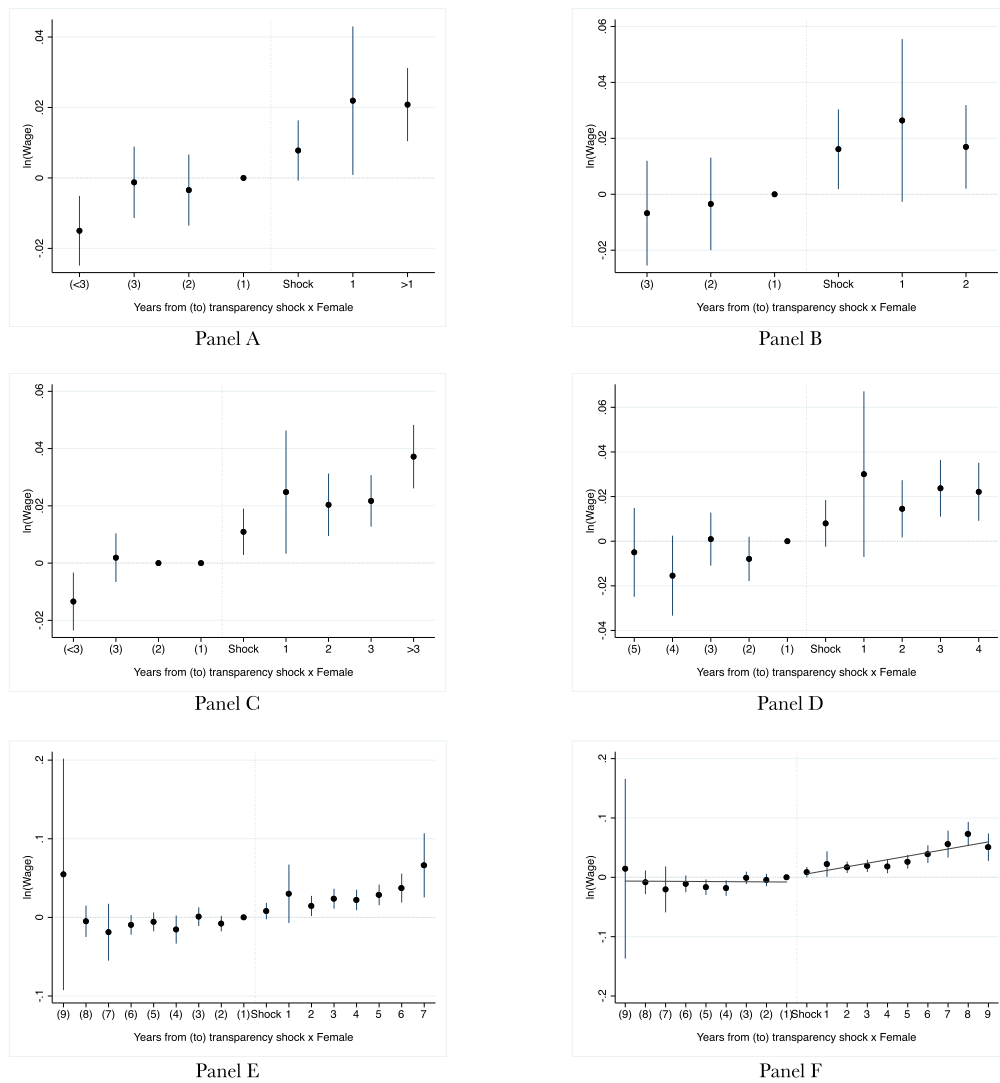
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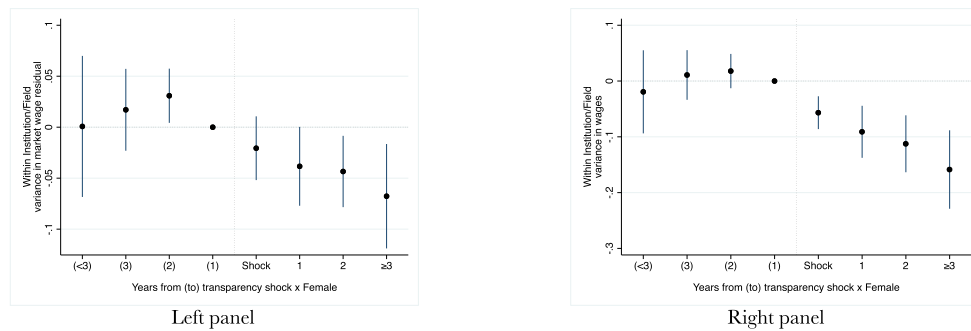
Extended Data Fig. 1 | The unconditional and conditional gender wage gap. Notes: The figure presents OLS regression estimates explaining (ln) salaries. Plotted coefficients of year dummies interacted with Female indicator, with 95% confidence intervals. Levels are scaled by the value on un-interacted Female indicator. Unconditional gap is based on a model with year dummies only. Conditional gap is based on models with year, academic domain, and institution fixed effects as well as controls for academic tenure (ln), number of academic articles, number of published books, number of awards, number of grants, and number of patents. Regression results used to generate this plot are reported in Supplementary Table 3.1.



Extended Data Fig. 2 | Equity in organizations: The effect of pay transparency on gender wage gap. Notes: The figure presents regression coefficients from an OLS regression model explaining (ln) wages. Reference category is 1 year prior to transparency shock. Plotted coefficients: dummy variable for Female interacted with years from (to) transparency shock with 95% CIs. Standard errors clustered on institution. Controls include academic tenure (ln), number of published academic articles, number of published books, number of awards, number of grants, and number of patents, and institution, individual, and year fixed effects. Regression results used to generate this plot are reported in Supplementary Table 3.3.

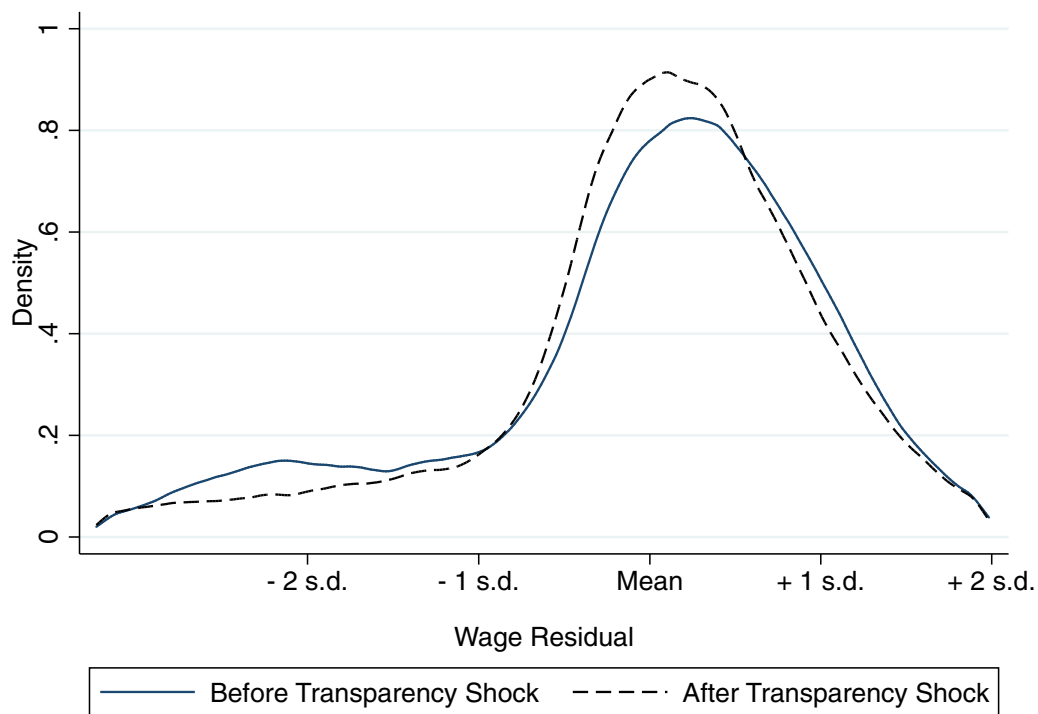


Extended Data Fig. 3 | Equity in organizations: The effect of pay transparency on gender wage gap: additional specifications. Notes: The figure presents regression coefficients from an OLS regression model explaining $\ln(\text{wage})$. Reference category is 1 year prior to transparency shock (1 and 2 years for 2SLS results, panel C). Plotted coefficients: dummy variable for Female interacted with years from (to) transparency shock with 95% CIs. Standard errors clustered on institution. Controls include academic tenure ($\ln$), number of published academic articles, number of published books, number of awards, number of grants, and number of patents, and institution, individual, and year fixed effects. Panel A – population restricted to 2004–2013 inclusive; Panel B – stacked difference in differences model (see text for more details). Panel C – 2SLS results, instrumented covariate: women's mean earnings as a % of men's in the private sector (see below for more details), excluded instrument: lead of transparency shock. Panel D and E – population restricted to CT, FL, PA, TX, VA (omitted states: California, New York, West Virginia). Panel F – full population. Regression results used to generate these plots are reported in Supplementary Tables 3.3, 3.4, and 3.5.



Extended Data Fig. 4 | Equity and equality in organizations: The effect of wage transparency on variance in market wage residuals and wage variance.

Notes: The figures present regression coefficients from an OLS regression model explaining variance in market wage residuals (Left panel) and variance in (ln) wages (Right panel). Reference category is 1 year prior to transparency shock. Both variables are calculated within Institution-Academic Field (11 categories). Plotted coefficients: years from (to) transparency shock with 95% CIs. Standard errors clustered on institution. Controls include reference group mean productivity levels and reference group productivity variances as well as year and academic field-institution fixed effects. Regression results used to generate these plots are reported in Supplementary Table 4.3.



Extended Data Fig. 5 | Equality in organizations: Distribution of market wage residuals, by transparency shock. Notes: The figure presents kernel density estimates of wage regression residuals by transparency shocks. Controls include institution, academic domain, and year fixed effects. Means and standard deviations are calculated averaging across all time periods. Residuals trimmed at 1% and 99%. Two-sample combined Kolmogorov-Smirnov tests for equality of distribution functions: 0.041, p -value < 0.001.

Extended Data Table 1 | Conditional gender wage gap by state and main academic disciplines

State / Academic Field	Conditional Gender Wage Gap (%)	
	2005	2015
California	13.4	1.1
Connecticut	19.3	7.6
Florida	8.0	5.8
New York	16.7	4.3
Pennsylvania	-1.4	3.0
Texas	12.8	5.9
Virginia	11.5	9.8
West Virginia	7.0	2.7
Humanities	9.7	2.4
Physical and Mathematical Sciences	13.6	4.4
Biological and Biomedical Sciences	18.3	4.2
Social and Behavioral Sciences	13.0	6.6
Engineering	7.0	6.4

Notes: The numbers presented in the table are based on regression estimates of conditional gender wage gap as specified in models reported in Extended Data Fig. 1. Models are run separately for each state or academic discipline. Positive values indicate female faculty underpaid compared to men faculty.

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Data Providers: Academics Analytics, FOIL Requests: State Comptrollers, Individual Public Universities

Data analysis

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All data that support the findings of this study along with statistical codes generating the results have been deposited in the Open Inter-university Consortium for Political and Social Research Repository (OPENICPSR) under project number 155541 and are available at: <https://doi.org/10.3886/E155541V1>. Names of individual academics and institutions have been blinded and are represented in the data with author-generated unique identifiers.

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Behavioural & social sciences study design

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Study description	This study examines longitudinal quantitative data involving the performance and pay of nearly all faculty from PhD granting institutions in eight US states before and after a shock to the transparency of pay.
Research sample	The sample includes essentially all academics from 139 public PhD granting institutions in eight US states (California, New York, Florida, Texas, Pennsylvania, Connecticut, Virginia, West Virginia) for which salary data can be obtained. We believe this sample represents essentially the entire population of academics in these 139 institutions, and that these institutions represent nearly all public PhD granting institutions in these states.
Sampling strategy	Our approach has been to obtain what amounts to essentially the population of academics from public PhD granting institutions in these eight states, rather than a sample of these individuals. These states represent more than 41% of the US population, but there could be an issue of how representative these institutions are of the broader US and world population of academics.
Data collection	Data were obtained from several archival sources. One key source of data is from a consulting firm that has aggregated performance data for all academics employed in PhD-granting institutions in the US. The other data sources are US state comptroller offices or individual academic institutions which have provided us salary data in response to Freedom of Information Act (FOIA) requests for data.
Timing	The data were all assembled in 2018-2019, but encompass archival data from 1997 to 2017.
Data exclusions	In general we did not exclude data except clearly described winsorizing, a common strategy dealing with salary data to account for outliers. As reported in the paper, we drop observations with missing data only when these data are required for the specific analysis performed. We also test all of our results with a restricted sample that has all data required for all analyses.
Non-participation	This question is not relevant for our analysis. All data was publicly available, though extraordinarily difficult to compile. Thus, there was no option to opt out or in.
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State number and/or type of variables recorded (e.g. correct button press, response time) and what statistics were used to establish that the subjects were performing the task as expected (e.g. mean, range, and/or standard deviation across subjects).

Acquisition

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Field strength

Specify in Tesla

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Diffusion MRI

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☐ Not used

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Statistic type for inference
(See [Eklund et al. 2016](#))

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- ☐ ☐ Graph analysis
- ☐ ☐ Multivariate modeling or predictive analysis

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